

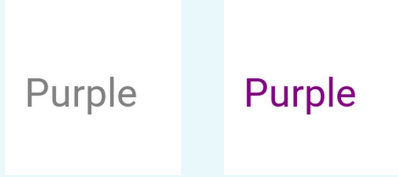
# Human Cognitive Control Tests

## Stroop Task

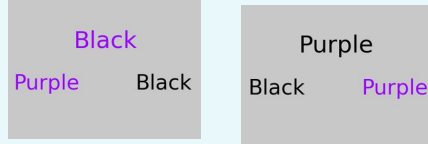
same pixel

Identify *color* while disregarding the word's meaning,

Stroop 1935,  
Smalker et al. 2022,  
Chen et al. 2016



Select the 2nd Row option whose *word meaning* matches the color of the 1st Row



## Flanker Task

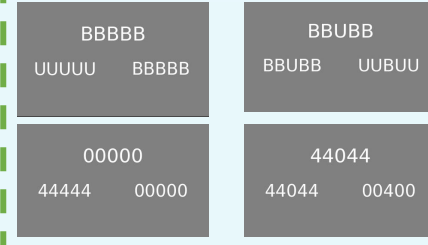
surrounding pixel

Identify *center* while disregarding surroundings,

Eriksen and  
Eriksen 1974



Select the 2nd Row option whose *center* matches 1st Row



Congruent

Incongruent

[Squared]  
Congruent

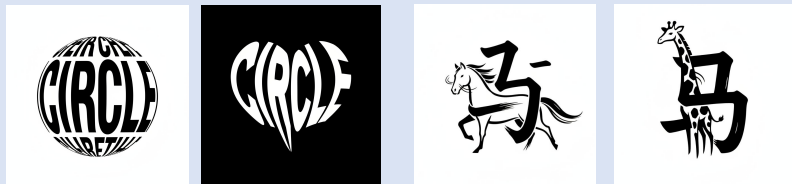
[Squared]  
Incongruent

Less complexity

Higher complexity

## OCR vs. Shape / Sketch

Identify *shape/text* while disregarding text/shape

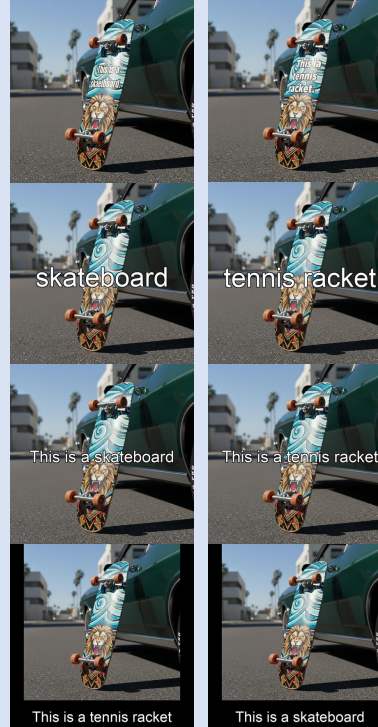


Identify *animal* disregarding animal name in Chinese

# Our Abstraction- and Modality-Aware Instantiations

## OCR vs. Recognition

Identify *object* while disregarding caption



Select the 2nd Row option whose *object* matches the caption of the 1st Row

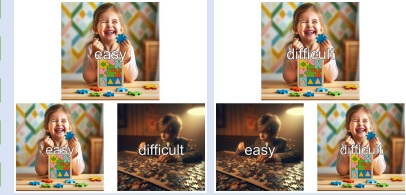


## OCR vs. Property



Identify *scenery property* while disregarding caption

\*prompt vary by property



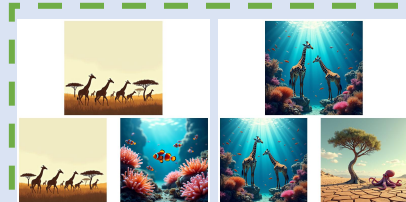
Select the 2nd Row option whose *scenery property* matches the caption of the 1st Row

## Visual Context vs. Knowledge

Creature-habitat knowledge



Identify *habitat* of animals while disregarding location in scene



Historical timeline knowledge



Identify *invention time* of tool disregarding time in scene

Select the 2nd Row option whose *scenery location* matches the habitat of the animals in the 1st Row